UNDERLYING SET OF THE INVERSE LIMIT OF AFFINE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. The closing primality verification is phrased 'It follows easily' in the source; it is a mechanical one-step check of a standard fact within the cited algebra framework, recorded here and kept verbatim in the span.

Allowed standard prerequisites (editorial).

- Primes of filtered colimits. Algebra, primes in filtered colimits of rings

Source reference. Original lemma 01YX: sources/stacks-chow/tags.txt, source lines 2572–2572. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $S = \lim S_i$ be the limit of a directed inverse system of schemes with affine transition morphisms (Lemma 01YX). Then $S_{set} = \lim_i S_{i,set}$ where S_{set} indicates the underlying set of the scheme S.*

Proof. Pick $i \in I$. Take $U_i \subset S_i$ an affine open. Denote $U_{i'} = f_{i'i}^{-1}(U_i)$ and $U = f_i^{-1}(U_i)$. Here $f_{i'i} : S_{i'} \to S_i$ is the transition morphism and $f_i : S \to S_i$ is the projection. By Lemma 01YX we have $U = \lim_{i' \geq i} U_i$. Suppose we can show that $U_{set} = \lim_{i' \geq i} U_{i',set}$. Then the lemma follows by a simple argument using an affine covering of S_i. Hence we may assume all S_i and S affine. This reduces us to the algebra question considered in the next paragraph.

Suppose given a system of rings $(A_i, \varphi_{ii'})$ over I. Set $A = \operatorname{colim}_i A_i$ with canonical maps $\varphi_i : A_i \to A$. Then

$$\operatorname{Spec}(A) = \lim_i \operatorname{Spec}(A_i)$$

Namely, suppose that we are given primes $\mathfrak{p}_i \subset A_i$ such that $\mathfrak{p}_i = \varphi_{ii'}^{-1}(\mathfrak{p}_{i'})$ for all $i' \geq i$. Then we simply set

$$\mathfrak{p} = \{x \in A \mid \exists i, x_i \in \mathfrak{p}_i \text{ with } \varphi_i(x_i) = x\}$$

It is clear that this is an ideal and has the property that $\varphi_i^{-1}(\mathfrak{p}) = \mathfrak{p}_i$. Then it follows easily that it is a prime ideal as well. □

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